Agile Project Management, Scrum, and Sprint Planning

Software Engineering & Project Management • Student Learning Book

Learning outcomes

• Explain Agile project management and how it differs from a plan-driven approach.

• Describe Scrum's accountabilities, events, and artefacts.

• Write a user story with clear acceptance criteria.

• Organize a product backlog and prioritize work based on value, risk, and dependencies.

• Plan a sprint using a Sprint Goal, team capacity, and a realistic set of backlog items.

• Use a simple Scrum workflow for a College Event Management System (CEMS).

Running case study: A college team is building CEMS. Students need to discover events, register, and receive confirmation, while organizers need to publish events and manage capacity.

1. What Is Agile Project Management?

Agile project management is an approach that delivers work in small increments, uses frequent feedback, and adapts plans as the team learns. Instead of assuming every requirement is known at the beginning, the team works with stakeholders to refine priorities and deliver useful outcomes iteratively.

Agile is not the absence of planning or documentation. It changes the timing and style of planning: plans are revisited as evidence and priorities change, while the team still manages scope, quality, risks, dependencies, and commitments.

Example: Rather than waiting until the end of a semester to release a complete event portal, a team may first deliver event browsing, then registration, then organizer tools, using feedback from students after each increment.

2. Agile and Plan-Driven Approaches

Dimension	Plan-driven approach (typical)	Agile approach (typical)
Planning	More detail established early	High-level direction plus frequent replanning
Delivery	Often larger delivery near the end	Small usable increments delivered regularly
Requirements	Change may require formal re-planning	Backlog can be reordered as needs evolve
Feedback	Often at milestones or acceptance	Frequent feedback during development
Progress evidence	Plan and milestone tracking	Working product, completed backlog items, and goals
Best fit	Stable scope, formal dependencies, or fixed approvals may favor it	Uncertain or evolving needs benefit from short feedback cycles

These are broad tendencies, not strict rules. Real projects often combine approaches. Agile does not guarantee faster or cheaper delivery; outcomes depend on team capability, scope, constraints, and stakeholder involvement.

3. Agile Values and Principles in Practice

The Agile Manifesto emphasizes individuals and interactions, working software, customer collaboration, and responding to change. The items on the right—processes and tools, comprehensive documentation, contract negotiation, and following a plan—still have value; the manifesto values the items on the left more highly.

• **Deliver value early:** release the smallest useful feature that solves a real problem.

• **Welcome useful feedback:** use feedback to improve the product and reprioritize work.

• **Collaborate frequently:** developers, users, and stakeholders clarify needs together.

• **Maintain quality:** testing, clear code, and sustainable pace help the team continue delivering.

• **Inspect and adapt:** review both the product and the team's working process.

4. Scrum Overview

Scrum is a lightweight framework for solving complex problems through iterative and incremental delivery. It defines accountabilities, events, and artefacts. Scrum is not a synonym for every Agile method; Kanban and Extreme Programming, for example, use different practices.

A Scrum Team is intended to be small enough to remain nimble and large enough to complete valuable work within a Sprint. The team is cross-functional and self-managing: it decides how to organize the work needed to meet the Sprint Goal.

Scrum accountabilities

Accountability	Main responsibility	CEMS example
Product Owner	Maximizes product value and manages the Product Backlog effectively	Orders work based on student and organizer needs
Scrum Master	Helps establish Scrum, improves team effectiveness, and helps address impediments	Facilitates events and helps remove a blocked test environment
Developers	Create a usable Increment each Sprint and plan/manage the work	Design, code, test, and integrate registration features

These are accountabilities, not necessarily separate job titles in every organization. A Scrum Master is not simply a task-assigning manager, and the Product Owner is not the person who unilaterally dictates every technical decision.

5. Scrum Events

Event	Purpose	Typical classroom illustration
Sprint	Fixed-length period, one month or less, to create a usable Increment	A two-week cycle to deliver event browsing and registration
Sprint Planning	Agree why the Sprint is valuable, what can be done, and how work will proceed	Choose a realistic set of backlog items and define the Sprint Goal
Daily Scrum	Developers inspect progress toward the Sprint Goal and adapt the plan	A short daily planning and coordination event
Sprint Review	Inspect the Increment with stakeholders and discuss what to do next	Students try registration and give feedback
Sprint Retrospective	Identify ways to improve quality and effectiveness	Team agrees to add earlier peer review next Sprint

A Sprint is the container for the other Scrum events. The Daily Scrum is for the Developers to inspect progress and adapt their plan; it is not intended to be a status report to a manager. Sprint Review and Retrospective have different purposes: product feedback versus improvement of how the team works.

6. Scrum Artefacts and Commitments

Artefact	What it contains	Commitment
Product Backlog	Ordered, evolving list of work needed to improve the product	Product Goal
Sprint Backlog	Sprint Goal, selected Product Backlog items, and the actionable plan	Sprint Goal
Increment	Usable, integrated product work that meets the Definition of Done	Definition of Done

The Product Goal describes a future state of the product. The Sprint Goal gives the Sprint a single coherent objective. The Definition of Done describes the quality state required for work to count as part of the Increment. A user story is a common way to express a backlog item, but Scrum does not require every backlog item to use that exact format.

7. Writing User Stories and Acceptance Criteria

A user story is a concise description of a capability from a user's perspective. A common template is:

> As a *[type of user]*, I want *[capability]*, so that *[benefit]*.

Example: As a student, I want to register for an available college event so that my place is reserved.

Acceptance criteria make expected behavior testable. For the story above:

• A signed-in student can register when seats are available.

• A successful registration shows a confirmation and unique registration reference.

• A student cannot register twice for the same event.

• When capacity is reached, the system prevents another confirmed registration and displays a clear message.

• The registration state is saved and can be retrieved after the user signs in again.

Acceptance criteria are not the same as a full test plan. They describe conditions that must be satisfied for the backlog item to be accepted; test cases specify particular inputs, steps, and expected results.

8. Product Backlog and Prioritization

The Product Backlog is an ordered list of product work. Items may be refined over time to clarify value, scope, dependencies, and acceptance criteria. Ordering should consider user value, risk reduction, urgency, dependencies, effort, and learning—not simply which stakeholder asks most loudly.

Backlog item	Value / reason	Example priority
Browse published events	Core student need; enables discovery	High
Register for an available event	Core product outcome	High
Prevent overbooking	Protects correctness and trust	High
Organizer creates an event	Enables event supply	High
Export attendance report	Useful for organizers, but may follow core registration	Medium
Animated landing-page effects	Visual enhancement, not essential to first release	Low

Priorities depend on context. For example, preventing overbooking may be a release blocker even if it appears less visible than a new screen. Dependencies and risk can change the order.

9. Estimation, Capacity, and Sprint Planning

Sprint Planning is a collaborative activity. The team considers the Sprint Goal, the highest-priority work, its understanding of the items, past performance where available, and actual capacity such as holidays, exams, support duties, and part-time availability.

• **Effort estimate:** an approximate assessment of the work involved.

• **Story points:** a relative estimate of complexity, uncertainty, and effort; they are not hours.

• **Capacity:** the team's available time and capability for the Sprint.

• **Velocity:** the amount of work completed in a Sprint, often measured in story points; useful as a planning aid, not an individual performance target.

• **Sprint Goal:** the outcome that gives the selected work a coherent purpose.

> **Illustration:** A team estimates its available capacity is lower during exam week. It selects event browsing and the basic registration flow, rather than also promising reporting, analytics, and visual redesign. The goal is a reliable registration journey, not maximizing the number of tickets.

10. Example Sprint Plan for CEMS

Item	Estimate (illustrative)	Acceptance focus
View published events	3 points	Student sees title, date, venue, and available status
View event details	2 points	Details page displays description and capacity status
Register for event	5 points	Successful registration creates a reference
Prevent duplicate registration	3 points	Second attempt is rejected clearly
Test registration limits	3 points	Capacity limit is respected under repeated attempts

Illustrative Sprint Goal: **Students can discover an event and complete one reliable registration when seats are available.** The estimate totals 16 points, but the team should select work based on its own historical delivery and actual capacity; the number is not a universal target.

11. A Simple Scrum Board

To Do	In Progress	Review / Test	Done
View event details Prevent duplicate registration	Build event list Implement registration API	Review acceptance criteria Run capacity tests	Definition of Done met Integrated and usable

A board visualizes workflow, but moving a card to Done should reflect the team's agreed Definition of Done. A task is not Done merely because coding has stopped; required integration, review, and testing must be completed.

12. Definition of Done and Increment Quality

A practical student-team Definition of Done might require:

• Acceptance criteria are satisfied.

• Code is reviewed by another team member.

• Relevant automated tests pass and important manual checks are recorded.

• Changes are integrated into the agreed branch.

• No known critical defect remains open for the item.

• User-facing or technical documentation is updated when needed.

Teams should agree on a Definition of Done that fits their context. It should be clear, visible, and applied consistently. Do not weaken it silently just to make a Sprint appear complete.

13. Scrum Risks and Common Mistakes

• Treating the Sprint as a mini-Waterfall phase with no feedback until the end.

• Overcommitting without considering team capacity or dependencies.

• Changing the Sprint Goal casually instead of discussing the impact with the team.

• Using story points to compare individual developers.

• Calling every meeting a Scrum event without using its purpose.

• Keeping a backlog as a fixed wish list instead of refining and ordering it.

• Demonstrating unfinished work as though it met the Definition of Done.

• Ignoring testing, accessibility, security, or technical debt to increase feature count.

14. Classroom Activity: Plan a Two-Week Sprint

In groups of four to six, plan the first CEMS Sprint.

1. Choose a Product Goal, such as enabling students to discover and register for events.
2. Create at least eight Product Backlog items and order them by value, risk, and dependencies.
3. Write two user stories with testable acceptance criteria.
4. Select a Sprint Goal and a realistic set of work based on your assumed team capacity.
5. Create a Sprint Backlog and a simple board with To Do, In Progress, Review/Test, and Done.
6. Define your Definition of Done and identify at least three risks or impediments.
7. Present the plan and explain what you deliberately left out of the Sprint.

> **Deliverables:** Product Backlog, two user stories with acceptance criteria, Sprint Goal, Sprint Backlog, board, Definition of Done, risk list, and a 3–5 minute explanation.

15. Review Questions with Suggested Answers

1. What makes Agile adaptive?

Suggested answer: It uses short delivery cycles, frequent feedback, and reprioritization based on learning.

2. Is Agile the same as Scrum?

Suggested answer: No. Agile is a broad set of values and approaches; Scrum is one framework.

3. Who is accountable for maximizing product value?

Suggested answer: The Product Owner.

4. What is the Scrum Master's role?

Suggested answer: Help establish Scrum, improve team effectiveness, and help address impediments.

5. What is the purpose of the Daily Scrum?

Suggested answer: Developers inspect progress toward the Sprint Goal and adapt their plan.

6. How does a Sprint Review differ from a Retrospective?

Suggested answer: Review inspects the product with stakeholders; Retrospective improves the team's way of working.

7. What is the Product Goal?

Suggested answer: The future state of the product that guides the Product Backlog.

8. What is a Sprint Goal?

Suggested answer: The single objective that gives the Sprint a coherent purpose.

9. Why should story points not be used to rank individual developers?

Suggested answer: They are relative estimates for planning work, not a reliable measure of individual productivity.

10. What is the Definition of Done?

Suggested answer: A shared description of the quality state required for an Increment to be considered complete.

16. Glossary

Term	Meaning
Agile	Adaptive approach emphasizing collaboration, feedback, and incremental value
Scrum	Lightweight framework for complex product work
Product Backlog	Ordered, evolving list of work to improve a product
Sprint	Fixed-length period of one month or less
User story	Concise user-perspective description of a capability
Acceptance criteria	Conditions that define expected behavior for a backlog item
Sprint Goal	Objective that provides focus and coherence during a Sprint
Increment	Usable, integrated product work meeting the Definition of Done

Term	Meaning
Velocity	Historical amount of work completed in a Sprint, used cautiously for planning
Retrospective	Event for inspecting and improving the team's way of working

17. Homework

1. Prepare a Product Backlog of at least ten items for a Library Management System or Hospital Appointment System.

2. Write three user stories and at least four acceptance criteria for each.

3. Create a Sprint Goal and select a realistic Sprint Backlog for a one- or two-week Sprint.

4. Explain the difference between Product Backlog, Sprint Backlog, and Increment.

5. Write a short reflection: which feedback would make you reorder the backlog, and why?

Next chapter: Day 26 continues project planning with documentation, project presentation, and practical use of Jira to track backlog items and team work.